

Computer Networking — Acronyms with Short Notes

1. Networking Basics

LAN — *Local Area Network*. A network confined to a small area such as an office or building, offering high speed and low cost.

MAN — *Metropolitan Area Network*. Spans a city or large campus, connecting multiple LANs, typically using fibre links.

WAN — *Wide Area Network*. Covers large geographical areas, often countries or continents; the Internet is the largest WAN.

OSI — *Open Systems Interconnection*. A 7-layer conceptual model (Physical to Application) defining how data communication should occur between systems.

TCP/IP — *Transmission Control Protocol / Internet Protocol*. A 4-layer protocol suite that is the practical foundation of the Internet.

PDU — *Protocol Data Unit*. The unit of data exchanged at a given OSI layer, e.g., frame, packet, segment.

NIC — *Network Interface Card*. Hardware that connects a computer to a network and provides a unique MAC address.

MAC — *Media Access Control*. A sublayer of the Data Link layer responsible for hardware addressing and channel access control.

IP — *Internet Protocol*. A Network layer protocol that handles logical addressing and routing of packets across networks.

ISO — *International Organization for Standardization*. The body that developed the OSI reference model and many global standards.

ITU — *International Telecommunication Union*. A UN agency that coordinates global telecommunication standards and radio spectrum allocation.

ARPA — *Advanced Research Projects Agency*. The US defence agency that funded ARPANET, the precursor to the modern Internet.

2. Signals & Modulation

NRZ — *Non-Return-to-Zero*. A line coding scheme where signal level stays constant for the duration of a bit, without returning to zero.

NRZI — *Non-Return-to-Zero Inverted*. A variant of NRZ where a transition (not the level) represents a binary 1.

FDM — *Frequency Division Multiplexing*. Multiplexing technique that divides the available bandwidth into separate frequency bands for each signal.

TDM — *Time Division Multiplexing*. Multiplexing technique that allocates distinct time slots to each signal on a shared channel.

WDM — *Wavelength Division Multiplexing*. Multiplexing of multiple optical signals on a fibre using different light wavelengths.

DWDM — *Dense Wavelength Division Multiplexing*. A high-capacity form of WDM that packs many closely spaced wavelengths onto a single fibre.

VHF — *Very High Frequency*. Radio frequency range of 30–300 MHz, used for FM radio and TV broadcasting.

UHF — *Ultra High Frequency*. Radio frequency range of 300 MHz–3 GHz, used for TV, Wi-Fi, and mobile networks.

IR — *Infrared*. Short-range wireless transmission using infrared light, e.g., remote controls.

RF — *Radio Frequency*. Electromagnetic wave frequencies used for wireless transmission of signals.

AM — *Amplitude Modulation*. Modulation technique where the carrier signal's amplitude varies with the message signal.

FM — *Frequency Modulation*. Modulation technique where the carrier signal's frequency varies with the message signal.

PM — *Phase Modulation*. Modulation technique where the carrier signal's phase varies with the message signal.

SNR — *Signal-to-Noise Ratio*. A measure comparing signal power to background noise power, indicating channel quality.

3. Transmission Technologies

PCM — *Pulse Code Modulation*. Converts analog signals to digital by sampling, quantizing, and encoding into binary form.

DSL — *Digital Subscriber Line*. Technology delivering high-speed digital data over ordinary copper telephone lines.

ADSL — *Asymmetric Digital Subscriber Line*. A form of DSL with higher download than upload speed, suited for home Internet.

GEO — *Geostationary Earth Orbit*. Satellite orbit (36,000 km) where the satellite appears fixed relative to Earth.

LEO — *Low Earth Orbit*. Satellite orbit (500–2000 km) offering low latency, used by constellations like Starlink.

MEO — *Medium Earth Orbit*. Satellite orbit (2000–36,000 km), commonly used for GPS and navigation satellites.

4. Data Link Layer

CRC — *Cyclic Redundancy Check*. An error-detection method using polynomial division to detect corrupted frames.

ARQ — *Automatic Repeat reQuest*. Error-control technique where receiver requests retransmission of corrupted or lost frames.

HDLC — *High-level Data Link Control*. A bit-oriented Data Link layer protocol used for point-to-point and multipoint links.

ALOHA — *Additive Links On-line Hawaii Area*. An early random-access protocol allowing stations to transmit anytime, with collisions resolved by retransmission.

CSMA — *Carrier Sense Multiple Access*. Protocol where a station senses the channel before transmitting to avoid collisions.

CSMA/CD — *CSMA / Collision Detection*. CSMA variant used in wired Ethernet that detects and handles collisions during transmission.

CSMA/CA — *CSMA / Collision Avoidance*. CSMA variant used in wireless LANs that avoids collisions before they occur.

LLC — *Logical Link Control*. Upper sublayer of the Data Link layer managing flow control and error checking.

FCS — *Frame Check Sequence*. A trailer field in a frame holding CRC bits used for error detection.

ACK — *Acknowledgement*. A control message confirming successful receipt of data.

NAK — *Negative Acknowledgement*. A control message indicating that data was not received correctly.

GBN — *Go-Back-N*. A sliding window ARQ protocol that retransmits all frames from the point of error onward.

SR — *Selective Reject / Selective Repeat*. An ARQ protocol that retransmits only the specific frames that were lost or corrupted.

IEEE — *Institute of Electrical and Electronics Engineers*. A professional body that defines many networking standards, notably the 802 series.

TDMA — *Time Division Multiple Access*. Channel access method assigning different time slots to multiple users on the same frequency.

FDMA — *Frequency Division Multiple Access*. Channel access method assigning different frequency bands to multiple users.

CDMA — *Code Division Multiple Access*. Channel access method where users share the same frequency but are separated by unique codes.

PPP — *Point-to-Point Protocol*. A Data Link protocol for direct connections between two nodes, e.g., dial-up links.

SLIP — *Serial Line Internet Protocol*. An older, simple protocol for transmitting IP packets over serial lines, largely replaced by PPP.

IEEE 802.3 — *Ethernet (CSMA/CD-based LAN)*. Standard defining wired Ethernet LANs using CSMA/CD.

IEEE 802.4 — *Token Bus LAN*. Standard defining a bus topology LAN using token-passing access control.

IEEE 802.5 — *Token Ring LAN*. Standard defining a ring topology LAN using token-passing access control.

IEEE 802.11 — *Wireless LAN (Wi-Fi)*. Standard defining wireless LAN communication, commonly known as Wi-Fi.

5. Network Layer Addressing

IPv4 — *Internet Protocol version 4*. 32-bit addressing scheme providing about 4.3 billion unique addresses.

IPv6 — *Internet Protocol version 6*. 128-bit addressing scheme designed to overcome IPv4 address exhaustion.

CIDR — *Classless Inter-Domain Routing*. Addressing method allowing flexible-length subnet masks instead of fixed classes, improving address use.

NAT — *Network Address Translation*. Technique that maps private IP addresses to a public IP for Internet access.

ICMP — *Internet Control Message Protocol*. Network layer protocol used for error reporting and diagnostics, e.g., ping.

6. Routing & Network Management

RIP — *Routing Information Protocol*. A distance-vector routing protocol using hop count as its metric.

OSPF — *Open Shortest Path First*. A link-state routing protocol that uses Dijkstra's algorithm to find shortest paths.

BGP — *Border Gateway Protocol*. The path-vector protocol used to route traffic between autonomous systems on the Internet.

IGRP — *Interior Gateway Routing Protocol*. Cisco's proprietary distance-vector protocol, predecessor to EIGRP.

EIGRP — *Enhanced Interior Gateway Routing Protocol*. An advanced Cisco distance-vector protocol offering faster convergence than IGRP.

TTL — *Time to Live*. A field limiting how many hops a packet can traverse before being discarded.

MTU — *Maximum Transmission Unit*. The largest packet size that can be transmitted over a given network link.

ARP — *Address Resolution Protocol*. Resolves a known IP address to its corresponding MAC address.

RARP — *Reverse Address Resolution Protocol*. Resolves a known MAC address to its corresponding IP address.

DHCP — *Dynamic Host Configuration Protocol*. Automatically assigns IP addresses and network configuration to hosts.

DNS — *Domain Name System*. Translates human-readable domain names into IP addresses.

AS — *Autonomous System*. A collection of IP networks under a single administrative routing policy.

IGP — *Interior Gateway Protocol*. Routing protocol used for routing within a single autonomous system, e.g., OSPF, RIP.

EGP — *Exterior Gateway Protocol*. Routing protocol used for routing between different autonomous systems, e.g., BGP.

VLSM — *Variable Length Subnet Mask*. Technique allowing subnets of different sizes within the same network for efficient IP use.

MPLS — *Multi-Protocol Label Switching*. High-performance technique that forwards packets using short labels rather than long IP lookups.

IGMP — *Internet Group Management Protocol*. Manages multicast group membership between hosts and routers.

PIM — *Protocol Independent Multicast*. A multicast routing protocol that works independently of the underlying unicast routing protocol.

QoS — *Quality of Service*. Mechanisms that prioritize and manage network traffic to meet performance requirements.

DSCP — *Differentiated Services Code Point*. A field in the IP header used to classify and prioritize traffic for QoS.

7. Transport Layer

TCP — *Transmission Control Protocol*. A connection-oriented, reliable Transport layer protocol with error and flow control.

UDP — *User Datagram Protocol*. A connectionless, unreliable but fast Transport layer protocol, suited for real-time applications.

SCTP — *Stream Control Transmission Protocol*. A message-oriented Transport protocol combining features of TCP and UDP, supporting multi-streaming.

MSS — *Maximum Segment Size*. The largest amount of data a TCP segment can carry, excluding headers.

RTT — *Round Trip Time*. The time taken for a signal to travel to a destination and back.

RTO — *Retransmission Time Out*. The time TCP waits before retransmitting an unacknowledged segment.

SACK — *Selective Acknowledgement*. TCP option allowing the receiver to acknowledge non-contiguous blocks of received data.

SYN — *Synchronize (TCP flag)*. TCP flag used to initiate and synchronize sequence numbers during connection setup.

FIN — *Finish (TCP flag)*. TCP flag used to gracefully terminate a connection.

RST — *Reset (TCP flag)*. TCP flag used to abruptly terminate an invalid or unwanted connection.

PSH — *Push (TCP flag)*. TCP flag instructing the receiver to deliver buffered data to the application immediately.

URG — *Urgent (TCP flag)*. TCP flag indicating that the segment contains urgent data needing priority processing.

CWND — *Congestion Window*. Sender-side TCP variable that limits the amount of data sent based on network congestion.

RWND — *Receiver Window*. Receiver-advertised TCP variable indicating available buffer space for flow control.

AIMD — *Additive Increase Multiplicative Decrease*. TCP congestion control strategy that increases the window slowly but cuts it sharply on loss.

ECN — *Explicit Congestion Notification*. Mechanism allowing routers to signal congestion without dropping packets.

8. Application Layer Protocols

HTTP — *HyperText Transfer Protocol*. Protocol for transferring web pages between clients and servers.

HTTPS — *HTTP Secure*. HTTP transmitted over an encrypted SSL/TLS connection for secure web communication.

FTP — *File Transfer Protocol*. Protocol for transferring files between client and server over a network.

SFTP — *Secure File Transfer Protocol*. A secure version of FTP that transfers files over an encrypted SSH connection.

TFTP — *Trivial File Transfer Protocol*. A simplified, connectionless file transfer protocol using UDP, often for booting devices.

SMTP — *Simple Mail Transfer Protocol*. Protocol used to send email messages between mail servers.

POP3 — *Post Office Protocol version 3*. Protocol that downloads email from a server to a local client, usually removing it from the server.

IMAP — *Internet Message Access Protocol*. Protocol that manages email directly on the server, allowing synchronization across devices.

SSH — *Secure Shell*. Protocol providing encrypted remote login and command execution over a network.

SSL — *Secure Sockets Layer*. An older cryptographic protocol for securing network communications, predecessor to TLS.

TLS — *Transport Layer Security*. The modern cryptographic protocol that secures data transmission, successor to SSL.

Telnet — *Teletype Network*. A protocol for unencrypted remote terminal access, now largely replaced by SSH.

NFS — *Network File System*. Protocol allowing files to be accessed over a network as if they were local.

SMB — *Server Message Block*. Protocol used for sharing files, printers, and resources on a network, common in Windows.

SNMP — *Simple Network Management Protocol*. Protocol used to monitor and manage devices on a network.

LDAP — *Lightweight Directory Access Protocol*. Protocol for accessing and maintaining distributed directory information, e.g., user accounts.

NTP — *Network Time Protocol*. Protocol used to synchronize clocks of computers over a network.

MIME — *Multipurpose Internet Mail Extensions*. Standard that allows email to carry non-text attachments like images and audio.

URL — *Uniform Resource Locator*. The address used to locate a specific resource on the Internet.

URI — *Uniform Resource Identifier*. A general identifier for a resource, of which a URL is one type.

HTML — *HyperText Markup Language*. The standard markup language used to create web pages.

CDN — *Content Delivery Network*. A distributed network of servers that delivers web content quickly based on user location.

9. Modern & Emerging Networking

IoT — *Internet of Things*. Network of physical devices embedded with sensors that communicate over the Internet.

SDN — *Software-Defined Networking*. Architecture that separates the network control plane from the data plane for centralized management.

NFV — *Network Functions Virtualization*. Replaces dedicated network hardware appliances with virtualized software functions.

MQTT — *Message Queuing Telemetry Transport*. A lightweight publish-subscribe messaging protocol designed for IoT devices.

CoAP — *Constrained Application Protocol*. A lightweight web transfer protocol designed for constrained IoT devices and networks.

M2M — *Machine-to-Machine*. Direct communication between devices without human intervention.

IaaS — *Infrastructure as a Service*. Cloud model providing virtualized computing infrastructure such as servers and storage.

PaaS — *Platform as a Service*. Cloud model providing a platform for developing and deploying applications.

SaaS — *Software as a Service*. Cloud model delivering software applications over the Internet on demand.

VM — *Virtual Machine*. A software emulation of a physical computer running its own operating system.

AMQP — *Advanced Message Queuing Protocol*. An open messaging protocol for reliable communication between applications and systems.

XMPP — *Extensible Messaging and Presence Protocol*. An XML-based protocol for real-time messaging and presence information.

5G — *5th Generation Mobile Network*. The latest mobile network standard offering very high speed and low latency.

LTE — *Long-Term Evolution*. A 4G standard for high-speed wireless communication for mobile devices.

BLE — *Bluetooth Low Energy*. A power-efficient variant of Bluetooth designed for short-range IoT communication.

LoRaWAN — *Long Range Wide Area Network*. A low-power, long-range wireless protocol designed for IoT applications.

NB-IoT — *Narrowband Internet of Things*. A low-power wide-area cellular technology designed for IoT devices.

VXLAN — *Virtual Extensible LAN*. A tunnelling protocol that extends Layer 2 networks over a Layer 3 infrastructure.

SD-WAN — *Software-Defined Wide Area Network*. Applies SDN principles to WANs for centralized, flexible traffic management.

VLAN — *Virtual Local Area Network*. A logically segmented LAN created within a physical network for isolation and management.

10. Organizations & Security

RFC — *Request for Comments*. Documents that define Internet standards, protocols, and best practices.

IETF — *Internet Engineering Task Force*. The body that develops and promotes Internet standards through RFCs.

IANA — *Internet Assigned Numbers Authority*. Organization responsible for coordinating IP addresses, domain names, and protocol parameters.

ICANN — *Internet Corporation for Assigned Names and Numbers*. Non-profit body overseeing domain name system and IP address allocation policy.

W3C — *World Wide Web Consortium*. The organization that develops standards for the World Wide Web, e.g., HTML, CSS.

ANSI — *American National Standards Institute*. Body that oversees development of voluntary standards in the United States.

EIA — *Electronic Industries Alliance*. Former US standards body for electronics, including cabling standards.

TIA — *Telecommunications Industry Association*. Develops standards for telecommunications equipment and cabling, often with EIA.

NIST — *National Institute of Standards and Technology*. US agency that develops measurement, cybersecurity, and technology standards.

ISP — *Internet Service Provider*. A company that provides individuals and organizations access to the Internet.

IXP — *Internet Exchange Point*. A physical location where ISPs interconnect to exchange traffic directly.

IDS — *Intrusion Detection System*. A system that monitors network traffic for suspicious activity and alerts administrators.

IPS — *Intrusion Prevention System*. A system that actively detects and blocks malicious network traffic in real time.

DMZ — *Demilitarized Zone*. A buffer subnetwork that exposes external-facing services while protecting the internal network.

ACL — *Access Control List*. Rules that permit or deny traffic on a network device based on defined criteria.

PAT — *Port Address Translation*. A form of NAT that maps multiple private addresses to one public IP using different ports.

VPN — *Virtual Private Network*. Creates a secure, encrypted tunnel for private communication over a public network.

SDU — *Service Data Unit*. Data passed down from an upper layer to a lower layer before encapsulation as a PDU.